

Two-Dimensional Heat Equation and Reaction–Diffusion Equation (100 points)

Consider the two-dimensional reaction–diffusion equation

$$\frac{\partial u}{\partial t} = k \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + u(10 - u), \quad 0 \leq x \leq 10, \quad 0 \leq y \leq 10,$$

where $k = 1$. The boundary conditions are

$$u(0, y, t) = 10, \quad u(10, y, t) = 5,$$

$$\frac{\partial u}{\partial y}(x, 0, t) = -q_0 \sin\left(\frac{\pi x}{10}\right), \quad \frac{\partial u}{\partial y}(x, 10, t) = q_0 \sin\left(\frac{\pi x}{10}\right),$$

where $q_0 = 0.2$. Let

$$\lambda = \frac{\pi}{10}.$$

The corresponding heat equation is obtained by omitting the reaction term $u(10 - u)$.

For the heat equation, use the initial condition

$$u(x, y, 0) = \sin\left(\frac{\pi x}{10}\right) \cos\left(\frac{\pi y}{10}\right) + 10 - 0.5x + \frac{q_0}{\lambda \sinh(5\lambda)} \sin(\lambda x) \cosh(\lambda(y - 5)).$$

For the reaction–diffusion equation, use the initial condition

$$u(x, y, 0) = 5 \sin\left(\frac{\pi x}{10}\right) + 10 - 0.5x + \frac{q_0}{\lambda \sinh(5\lambda)} \sin(\lambda x) \cosh(\lambda(y - 5)).$$

For all numerical simulations, solve the problem over

$$0 \leq t \leq t_f, \quad t_f = 0.5.$$

Unless otherwise specified, use the same time step Δt for different time-integration methods so that the numerical results can be compared directly.

1. For the heat equation

$$\frac{\partial u}{\partial t} = k \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right),$$

derive the analytical solution corresponding to the initial condition

$$u(x, y, 0) = \sin\left(\frac{\pi x}{10}\right) \cos\left(\frac{\pi y}{10}\right) + 10 - 0.5x + \frac{q_0}{\lambda \sinh(5\lambda)} \sin(\lambda x) \cosh(\lambda(y - 5)).$$

Verify that the solution satisfies the boundary conditions. (5 points)

2. Discretize the computational domain using a uniform grid with spacing

$$h = \frac{10}{N}.$$

The grid points are

$$x_i = ih, \quad y_j = jh, \quad i, j = 0, 1, \dots, N.$$

Because the boundary values at $x = 0$ and $x = 10$ are prescribed, the unknowns are located at

$$i = 1, 2, \dots, N-1, \quad j = 0, 1, \dots, N.$$

Define the unknown vector $\vec{u}$ and clearly describe the index ordering used to map $u_{i,j}$ to a one-dimensional vector. (5 points)

3. Discretize the spatial derivatives using second-order central differences. Apply the Dirichlet boundary conditions in the x -direction and the Neumann boundary conditions in the y -direction.

Construct the discrete Laplacian matrix A and boundary contribution vector $\vec{b}$ such that

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \approx A\vec{u} + \vec{b}.$$

For $N = 10$, describe the size and sparse structure of A , and explicitly derive how $\vec{b}$ incorporates the prescribed Dirichlet boundary values at $x = 0$ and $x = 10$, as well as the non-homogeneous Neumann boundary conditions at $y = 0$ and $y = 10$. (10 points)

4. Discretize the heat equation in time using the explicit Euler method:

$$\vec{u}^{n+1} = \vec{u}^n + \Delta t k \left(A\vec{u}^n + \vec{b} \right).$$

Derive the stability condition for the explicit Euler method using eigenvalue-based stability analysis of the matrix A . Choose a stable time step for each value of

$$N = 10, 20, 40, 80.$$

For example, you may use

$$\Delta t = 0.1h^2.$$

(10 points)

5. For the heat equation, compute numerical solutions using the explicit Euler method for

$$N = 10, 20, 40, 80.$$

Compare the numerical solutions with the analytical solution at $t = t_f$. Plot the heat-equation solution distributions for each N at

$$t = 0, 0.1, 0.2, 0.3, 0.4, 0.5.$$

(10 points)

6. Discretize the heat equation in time using the implicit Euler method:

$$\frac{\vec{u}^{n+1} - \vec{u}^n}{\Delta t} = k \left(A \vec{u}^{n+1} + \vec{b} \right).$$

Write the resulting linear system in the form

$$B \vec{u}^{n+1} = \vec{u}^n + \Delta t k \vec{b}.$$

Define the matrix B . For $N = 10$, describe the size and sparse structure of B . (10 points)

7. Compute numerical solutions using the implicit Euler method for

$$N = 10, 20, 40, 80.$$

Plot the implicit Euler solutions at

$$t = 0, 0.1, 0.2, 0.3, 0.4, 0.5.$$

Compare these plots with the explicit Euler solutions.

(10 points)

8. Compute the relative errors of both the explicit Euler and implicit Euler methods in the discrete L_2 -norm at $t = t_f$:

$$E_{L_2} = \frac{\left(\sum_{i,j} |u_{i,j}^{\text{num}} - u(x_i, y_j, t_f)|^2 \right)^{1/2}}{\left(\sum_{i,j} |u(x_i, y_j, t_f)|^2 \right)^{1/2}}.$$

Generate a log-log plot of E_{L_2} versus h , and discuss the observed convergence rate. (5 points)

9. Compute converged numerical solutions using the implicit Euler method with a large time step for

$$N = 10, 20, 40, 80.$$

Continue the iteration until the solution is nearly converged, for example until

$$\frac{\|\vec{u}^{n+1} - \vec{u}^n\|_2}{\|\vec{u}^{n+1}\|_2} < 10^{-8}.$$

Compare the numerical steady-state solution with the analytical steady-state solution

$$u_\infty(x, y) = 10 - 0.5x + \frac{q_0}{\lambda \sinh(5\lambda)} \sin(\lambda x) \cosh(\lambda(y - 5)).$$

(5 points)

10. For the reaction–diffusion equation

$$\frac{\partial u}{\partial t} = k \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) + u(10 - u),$$

use a semi-implicit Crank–Nicolson method in which the diffusion term is treated by the Crank–Nicolson method and the reaction term is treated explicitly:

$$\frac{\vec{u}^{n+1} - \vec{u}^n}{\Delta t} = \frac{1}{2}kA \left(\vec{u}^{n+1} + \vec{u}^n \right) + k\vec{b} + \vec{u}^n \circ (10 - \vec{u}^n).$$

Rearrange the method into the linear system

$$\left(I - \frac{\Delta t}{2}kA \right) \vec{u}^{n+1} = \left(I + \frac{\Delta t}{2}kA \right) \vec{u}^n + \Delta t k\vec{b} + \Delta t \vec{u}^n \circ (10 - \vec{u}^n).$$

Define the matrix

$$B = I - \frac{\Delta t}{2}kA.$$

For $N = 10$, describe the size and sparse structure of B . Explain why this method avoids solving a nonlinear system at each time step.

For the reaction–diffusion simulations, use

$$\Delta t = 0.1h^2.$$

(10 points)

11. Implement the semi-implicit method for the reaction–diffusion equation using

$$N = 10, 20, 40, 80.$$

Discuss the effect of diffusion and nonlinear growth on the solution. Plot the reaction–diffusion solution distributions for each N at

$$t = 0, 0.1, 0.2, 0.3, 0.4, 0.5.$$

(10 points)

12. Compare the computational behavior of the three approaches used in this project:

explicit Euler, implicit Euler, semi-implicit Crank–Nicolson method.

Discuss stability, accuracy, computational cost, and implementation complexity. (10 points)